White Paper: The ChronAeon TimeTree (`chronaeon.timetree`) Algorithm and Coalescent Skyline Dynamics

Authors: ChronAeon & Phylodate Benchmark Working Group
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Abstract

We present the mathematical formulation, algorithmic design, and demographic skyline evaluation of **ChronAeon TimeTree** (`chronaeon.timetree`), a hybrid distance-geometry and tree-isotonic solver that bridges tree-free molecular clock inference with full time-calibrated phylogenetic reconstruction. Unlike conventional post-hoc dating tools in `phylodate` (`lsd2`, `treetime`, `treedater`, `reltime`, `node_dating`, `wlogdate`, `chronumental`, `chronos`, `treepl`, `mdcat`)—which require an external maximum-likelihood substitution phylogeny (95–99.9% of total compute time)—and unlike baseline scalar **ChronAeon Mode 1**—which estimates only $(\hat{\mu}, \hat{t}_{\text{MRCA}})$ and incurs a positive ancestral profile self-diversity offset $+D(R, Q(\gamma^*))$ on small trees—`chronaeon.timetree` recovers a complete, temporally consistent binary time-tree directly from either raw nucleotide sequences (`chronaeon_timetree_seq`) or substitution trees (`chronaeon_timetree_tree`) in 5–900 ms. Furthermore, we derive and benchmark the exact mathematical mechanisms governing how **heterochronous coalescent skyline plots** $(\widehat{N}_e(t))$ and **Lineages-Through-Time (LTT)** curves differ between `chronaeon.timetree` and `phylodate` timetrees across synthetic demographic trajectories (constant, exponential growth, and epidemic boom-bust) and empirical viral outbreaks (Rabies, Ebola, Chikungunya, and Dengue), and introduce closed-form + ensemble **95% confidence intervals for $N_e(t)$ over time** (`ChronAeonTimeTree.skyline(ci=True)`), achieving 92.5–98.1% empirical coverage (mean 94.8%) of ground-truth trajectories without MCMC.

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1. Motivation: Bridging Continuous Distance Geometry and Discrete Time-Trees

Molecular clock dating methods historically divide into two paradigms:

1. **Two-Stage Substitution-Tree Calibration (`phylodate`)**: A maximum-likelihood (ML) unrooted phylogeny $T_u = (V, E, \mathbf{b})$ with branch lengths b_e (in substitutions/site) is first inferred via IQ-TREE or RAxML, after which internal node timestamps $\mathbf{t}_I \in \mathbb{R}^{|V_I|}$ and clock rate μ are estimated via constrained weighted least squares (`lsd2`, `node_dating`), penalized/Gamma likelihood (`treedater`, `chronos`, `treepl`), log-transformed rate regularization (`wlogdate`), differentiable optimization (`chronumental`), or iterative marginal message passing (`treetime`).
2. **Tree-Free Continuous Distance Geometry (`chronaeon` Mode 1)**: Given an alignment $\mathbf{S} = \{s_1, \dots, s_N\} \in \{A, C, G, T\}^{N \times L}$ with tip sampling times $\mathbf{t} = (t_1, \dots, t_N)^\top$, ChronAeon constructs a continuous time-decay soft ancestral nucleotide profile $Q(\gamma^*) \in \Delta^{L \times 4}$ and regresses analytical Tamura-Nei (TN93) divergences $d_i(\gamma^*) = D_{\text{TN93}}(s_i, Q(\gamma^*))$ against t_i.

Each paradigm exhibits a distinct structural limitation: - **ML Tree Bottleneck & Root Sensitivity in `phylodate`**: Inferring T_u requires $O(N^2 L)$ likelihood evaluations (10^1–10^3 s), and downstream dating methods (`lsd2`, `treedater`, `reltime`, `wlogdate`) degrade severely if supplied an unrooted ML tree without explicit root optimization or when zero-mutation short branches ($b_e = 0$) attenuate the global clock rate $\hat{\mu}$. - **The Profile Self-Diversity Offset in Uncentered Scalar ChronAeon**: On an additive tree with true root R, the expected distance from tip i to the convex combination $Q(\gamma) = \sum_{j=1}^{N} w_j(\gamma) s_j$ satisfies:

$$\mathbb{E}[D(s_i, Q(\gamma))] = d(R, s_i) + \underbrace{\sum_{j=1}^{N} w_j(\gamma) d(R, s_j)}_{+\, D(R, Q(\gamma)) > 0} - 2 \sum_{j=1}^{N} w_j(\gamma) d(R, \text{MRCA}(i, j)). \tag{1}$$

Because the earliest tips lie at positive evolutionary distance from the true root R, the constant term $+D(R, Q(\gamma)) > 0$ inflates the raw regression intercept d_0 and shifts the uncentered scalar estimate $\hat{t}_{\text{MRCA}} = \bar{t} - \bar{d}/\hat{\mu}$ backward in

time on small cohorts ($N < 50$).

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2. The ChronAeon TimeTree (`chronaeon.timetree`) Algorithm

`chronaeon.timetree` resolves both limitations through a six-stage analytical pipeline implemented in `chronaeon/src/chronaeon/timetree.py`.

Stage 1: Time-Decay Soft-Profile Construction & The Centering Identity

Given tip timestamps $\mathbf{t} \in \mathbb{R}^N$ with $t_{\min} = \min_i t_i$ and $\bar{t} = \frac{1}{N}\sum_i t_i$, we define normalized exponential time-decay weights over a discrete candidate grid $\gamma \in \{0, 0.25, 0.5, 1, 2, 4, 7, 10\}$:

$$w_i(\gamma) = \frac{\exp\left(-\gamma \frac{t_i - t_{\min}}{\max(\bar{t} - t_{\min}, 10^{-5})}\right)}{\sum_{k=1}^{N} \exp\left(-\gamma \frac{t_k - t_{\min}}{\max(\bar{t} - t_{\min}, 10^{-5})}\right)}, \qquad Q_{\ell,b}(\gamma) = \sum_{i=1}^{N} w_i(\gamma)\, \mathbb{I}(s_{i,\ell} = b). \tag{2}$$

In sequence mode, $d_{\text{prof},i}(\gamma) = D_{\text{TN93}}(s_i, Q(\gamma))$; in distance/tree mode, we evaluate the exact quadratic distance-space dual:

$$d_{\text{prof},i}(\gamma) = \sum_{j=1}^{N} w_j(\gamma) D_{ij} - \frac{1}{2}\sum_{j=1}^{N}\sum_{k=1}^{N} w_j(\gamma) w_k(\gamma) D_{jk}. \tag{3}$$

Selecting $\gamma^* = \arg\max_\gamma \text{Corr}(\mathbf{t}, \mathbf{d}_{\text{prof}}(\gamma))$ yields the optimal soft profile. Crucially, subtracting the sample mean $\bar{d}_{\text{prof}} = \frac{1}{N}\sum_{i=1}^{N} d_{\text{prof},i}(\gamma^*)$ forms the **centered profile divergence vector**:

$$\tilde{\mathbf{d}}_{\text{prof}} = \mathbf{d}_{\text{prof}}(\gamma^*) - \bar{d}_{\text{prof}}\mathbf{1}, \tag{4}$$

which **identically cancels** the constant ancestral offset $+D(R, Q(\gamma^*))$ in Eq. (1) while preserving the relative root-to-tip evolutionary displacements across all taxa.

Stage 2: Fast Rust TN93 Distance Manifold & Vectorized Neighbor-Joining

- **Tree-Free Mode (`chronaeon_timetree_seq`)**: The native Rust kernel (`aeon_core._native.tn93_distance_matrix`) computes the $N \times N$ Tamura-Nei distance matrix $\mathbf{D}$ in parallel across sites. An unrooted binary tree $T_u = (V, E, \mathbf{b})$ with $2N-2$ nodes and $2N-3$ edges is reconstructed in $O(N^3)$ via vectorized Neighbor-Joining (`fast_nj`) using the Q-criterion matrix:

$$Q_{uv} = (m-2)D_{uv} - \sum_{k=1}^{m} D_{uk} - \sum_{k=1}^{m} D_{vk}, \qquad b_{uw} = \frac{1}{2}D_{uv} + \frac{r_u - r_v}{2(m-2)}. \tag{5}$$

- **Tree-Guided Mode (`chronaeon_timetree_tree`)**: Parses an unrooted or rooted Newick tree directly into an adjacency representation and extracts the all-nodes-to-tips patristic distance matrix in $O(N^2)$.

Stage 3: Closed-Form Soft-Profile-Guided Continuous Edge Rooting

For every candidate edge $e = (u, v) \in E$ of length b_e and fractional split $\alpha \in [0, 1]$ (placing root $R_{e,\alpha}$ at distance αb_e from u and $(1-\alpha)b_e$ from v), the induced root-to-tip patristic distance vector across tips $i \in \{1, \ldots, N\}$ is linear in α:

$$\mathbf{d}(e, \alpha) = \mathbf{d}_0(e) + \alpha b_e \mathbf{s}_e, \qquad s_{e,i} = \begin{cases} +1 & \text{if tip } i \text{ is in the } u\text{-subtree,} \\ -1 & \text{if tip } i \text{ is in the } v\text{-subtree.} \end{cases} \tag{6}$$

Let $\tilde{\mathbf{x}} = \mathbf{x} - \bar{x}\mathbf{1}$ denote mean-centering and $\mathbf{M}_t = \mathbf{I} - \frac{\tilde{\mathbf{t}}\tilde{\mathbf{t}}^\top}{\|\tilde{\mathbf{t}}\|_2^2}$ denote the orthogonal projector onto $\text{span}(\mathbf{1}, \mathbf{t})^\perp$. Both the temporal-regression residual minimizer $\alpha^*_{\text{rss}}(e)$ and the centered soft-profile alignment minimizer $\alpha^*_{\text{prof}}(e)$ admit exact closed-form solutions on $[0, 1]$:

$$\alpha^*_{\text{rss}}(e) = \text{clip}_{[0,1]}\left(-\frac{\langle \mathbf{M}_t\tilde{\mathbf{d}}_0(e),\, \mathbf{M}_t\tilde{\mathbf{s}}_e\rangle}{b_e\|\mathbf{M}_t\tilde{\mathbf{s}}_e\|_2^2}\right), \qquad \alpha^*_{\text{prof}}(e) = \text{clip}_{[0,1]}\left(\frac{\langle \tilde{\mathbf{d}}_{\text{prof}} - \tilde{\mathbf{d}}_0(e),\, \tilde{\mathbf{s}}_e\rangle}{b_e\|\tilde{\mathbf{s}}_e\|_2^2}\right). \tag{7}$$

Evaluating the composite objective over all $2N-3$ edges subject to positive evolutionary rate $\hat{\mu}(e,\alpha) = \langle \tilde{\mathbf{t}}, \tilde{\mathbf{d}}(e,\alpha)\rangle / \|\tilde{\mathbf{t}}\|_2^2 > 0$:

$$\mathcal{J}(e,\alpha) = \left\|\tilde{\mathbf{d}}(e,\alpha) - \hat{\mu}(e,\alpha)\tilde{\mathbf{t}}\right\|_2^2 + \lambda_{\text{prof}}\left\|\tilde{\mathbf{d}}(e,\alpha) - \tilde{\mathbf{d}}_{\text{prof}}\right\|_2^2 \qquad (\lambda_{\text{prof}} = 0.35) \tag{8}$$

selects the globally optimal root (e^*, α^*) in $O(N^2)$ total operations.

Stage 4: Rank-1 Sherman-Morrison LOOCV Outlier Downweighting

Before solving for internal node dates, high-leverage sequencing errors, recombinants, or passaged lab strains are screened in $O(N)$ using analytical leave-one-out cross-validation (LOOCV) without refitting:

$$h_{ii} = \frac{1}{N} + \frac{(t_i - \bar{t})^2}{\sum_{j=1}^{N}(t_j - \bar{t})^2}, \qquad \hat{\sigma}^2_{(-i)} = \frac{1}{N-3}\left(\text{RSS} - \frac{r_i^2}{1-h_{ii}}\right), \qquad Z_i = \frac{r_i}{\hat{\sigma}_{(-i)}\sqrt{1-h_{ii}}}. \tag{9}$$

Taxa exceeding $|Z_i| > z_{\text{thresh}}$ (3.0 by default) receive Huber-type terminal branch weights $u_i = \min(1, (z_{\text{thresh}}/|Z_i|)^2)$.

Stage 5: Active-Set Contracted Tree-Laplacian Isotonic WLS Solver

Directed from R_{e^*,α^*}, the rooted topology has N fixed terminal timestamps $t_1, \ldots, t_N$ and $N-1$ unknown internal node timestamps $\mathbf{t}_I \in \mathbb{R}^{N-1}$. Under a Poisson-regularized branch variance model, each directed edge $e = (p \to v)$ carries weight $w_e = \frac{L_{\text{eff}}}{b_e + c/L_{\text{eff}}}$ ($c = 10$, scaled by u_v on terminal edges). We minimize the root-anchored isotonic WLS objective:

$$\min_{\mathbf{t}_I \in \mathbb{R}^{N-1},\, \mu > 0} \sum_{e=(p\to v)\in E} w_e\left(b_e - \mu(t_v - t_p)\right)^2 + w_{\text{anchor}}\left(t_{\text{root}} - t^{\text{rtt}}_{\text{root}}(\mu)\right)^2 \quad \text{subject to} \quad t_p \le t_v \;\; \forall (p \to v) \in E, \tag{10}$$

where $t^{\text{rtt}}_{\text{root}}(\mu) = \bar{t} - \bar{d}_{\text{root}}/\mu$ is the centered root-to-tip intercept. 1. **Active-Set Union-Find Contraction ($O(N)$ per iteration)**: For fixed μ, any violated internal edge $(p \to c)$ with $t_p > t_c$ is added to the active constraint set $\mathcal{A}_I$ and contracted via Union-Find into a connected component $C_k \in \mathcal{C}$, while any internal node exceeding a child tip ($t_p > t_{\text{leaf}}$) pins its component to $\min_{\text{leaf}\in\partial C_k} t_{\text{leaf}}$. The remaining free components satisfy a sparse, symmetric positive-definite **contracted tree-Laplacian linear system**:

$$\mathbf{L}_{\mathcal{C}}\mathbf{t}_{\mathcal{C}} = \mathbf{r}_{\mathcal{C}}(\mu), \tag{11}$$

which is solved in $O(|\mathcal{C}|) \subseteq O(N)$ time via sparse Cholesky/LU (`scipy.sparse.linalg.spsolve`). 2. **Blended Rate Update (Preventing Polytomy Attenuation)**: When short internal branches have $b_e = 0$ but $\tau_e = t_v - t_p > 0$, pure branch-ratio WLS updates ($\mu_{\text{wls}} = \frac{\sum w_e b_e \tau_e}{\sum w_e \tau_e^2}$) are biased downward by zero-mutation denominators. `chronaeon.timetree` blends the path-wise root-to-tip rate $\hat{\mu}_{\text{rtt}}$ with the branch-wise WLS/Poisson MLE rate $\hat{\mu}_{\text{branch}} = 0.7\hat{\mu}_{\text{wls}} + 0.3\frac{\sum b_e}{\sum \tau_e}$:

$$\hat{\mu}^{(k+1)} = \frac{1}{2}\hat{\mu}_{\text{rtt}} + \frac{1}{2}\hat{\mu}^{(k)}_{\text{branch}}, \tag{12}$$

guaranteeing immunity to internal polytomy rate collapse.

Stage 6: Dispersion-Calibrated Analytical Fieller Confidence Intervals

To account for phylogenetic non-independence and relaxed/autocorrelated rate drift without bootstrap resampling, `chronaeon.timetree` computes: 1. **Bartlett's Effective Sample Size**: $N_{\text{eff}} = \text{clip}_{[3,N]}\left(\frac{N^2}{\sum_{i,j}\exp(-\kappa D_{ij})}\right)$ with $\kappa = 5/\text{median}(D_{ij})$ and effective degrees of freedom $\nu^* = \max(2, N_{\text{eff}} - 2)$. 2. **Path-to-Branch Overdispersion Ratio**: Comparing path-level Poisson dispersion $\hat{\phi}_{\text{rtt}} = \frac{1}{N-2}\sum_i \frac{u_i L(d_i - \hat{\mu}(t_i - \hat{t}_{\text{root}}))^2}{\max(1/L, \hat{\mu}(t_i - \hat{t}_{\text{root}}))}$ against branch-level WLS dispersion $\hat{\phi}_{\text{branch}} = \frac{1}{|E|-|V_I|}\sum_e w_e(b_e - \hat{\mu}\tau_e)^2$ yields the autocorrelation inflation factor $\rho_{\text{disp}} = \sqrt{\max(1, \hat{\phi}_{\text{rtt}}/\hat{\phi}_{\text{branch}})}$. 3. **Exact Non-Linear Fieller Bounds**: Defining $g = \left(\frac{t_{0.975,\nu^*}\text{SE}_{\text{eff}}(\hat{\mu})}{\hat{\mu}}\right)^2$, when $g < 1$, the 95% confidence interval for t_{MRCA} (and analogously for every internal node $u \in V_I$ relative to its descendant tips) is solved in closed form via Fieller's quadratic root formula:

$$\text{CI}_{95\%}(t_{\text{MRCA}}) = \bar{t} - \frac{1}{1-g}\left[\hat{\theta}_0 \mp \sqrt{\hat{\theta}_0^2 - (1-g)\left(\hat{\theta}_0^2 - \frac{t^2_{0.975,\nu^*}\text{SE}^2(\hat{d}_0)}{\hat{\mu}^2}\right)}\right], \qquad \hat{\theta}_0 = \bar{t} - \hat{t}_{\text{MRCA}}. \tag{13}$$

3. How Coalescent Skyline Plots Differ Between `phylodate` Timetrees and `chronaeon.timetree`

Given a dated heterochronous phylogeny with N tip sampling times $s_1 \leq ... \leq s_N$ and $N-1$ internal coalescent times $c_1 \leq ... \leq c_{N-1}$, the sorted timeline of all $2N-1$ events partitions $[c_1, s_N]$ into elementary intervals $I_j = [t_{j-1}, t_j]$ of duration $\Delta t_j = t_j - t_{j-1} \geq 0$ with k_j extant lineages. Under the heterochronous Kingman coalescent (Rodrigo & Felsenstein 1999; Strimmer & Pybus 2001), the generalized skyline estimator over a pooled window W_m containing $r_m \geq 1$ coalescent events and spanning $\sum_{j \in W_m} \Delta t_j \geq \epsilon$ is:

$$\hat{N}_e(W_m) = \frac{1}{r_m} \sum_{j \in W_m} \binom{k_j}{2} \Delta t_j. \tag{14}$$

Because $\hat{N}_e(t)$ is a linear functional of the **inter-node waiting times Δt_j weighted quadratically by the extant lineage count** $\binom{k_j}{2}$, differences in how `chronaeon.timetree` and the `phylodate` methods estimate internal node dates manifest in four distinct ways on skyline and LTT plots:

3.1 The 1-Mutation Temporal Resolution Quantum and Zero-Wait Polytomy Collapse

On an alignment of length L evolving at substitution rate μ, a single mutation along a lineage corresponds to an expected waiting time of:

$$\Delta t_{\text{quantum}} = \frac{1}{\mu L}. \tag{15}$$

During rapid epidemic transmission (where k_j is large and true coalescent waiting times satisfy $\Delta t_j \ll (\mu L)^{-1}$), most short internal branches accumulate zero mutations ($b_e = 0$). Different dating algorithms handle this regime differently: - **Hard Topological Collapsing (`lsd2`)**: By default, `lsd2` collapses all internal branches with $b_e \leq 0.5/L$ (`nullblen = 0.5/L`) into multifurcations and clamps active constraints $t_p = t_v$. On empirical outbreaks, `lsd2` collapses **85.1% of coalescent intervals on Ebola**, **51.9% on Chikungunya Brazil**, and **46.7% on Rabies** to $\Delta t_j = 0$ (Figure 3B). In an ungrouped classic skyline ($\epsilon \to 0$), these zero-duration intervals produce artificial $\hat{N}_e(t) \to 0$ downward spikes; even with ϵ-pooling, if multifurcating nodes of out-degree $d > 2$ are not expanded with multiplicity $d-1$, the LTT curve $L(t)$ undercounts active lineages and biases $\binom{k_j}{2}$ upward by 3–10×. - **Active-Set Tree-Laplacian with Root-to-Tip Rate Blending (`chronaeon.timetree`)**: Both `chronaeon_timetree_tree` and `chronaeon_timetree_seq` enforce exact monotonicity $t_p \leq t_v$ via Union-Find active-set contraction while preserving the full binary topology ($N-1$ internal nodes) and preventing zero-length branches from attenuating $\hat{\mu}$. Consequently, `chronaeon.timetree` exhibits substantially lower zero-wait discretization than `lsd2` (**14.8%–18.5% on Chikungunya**, **24.4%–31.1% on Rabies**, and **68.6%–74.7% on ultra-low-diversity Ebola**; Figure 3B). - **Strict Positive Branch Regularization (`treedater`, `wlogdate`)**: Because `treedater` (Gamma-Poisson branch prior) and `wlogdate` ($\sum_e (\log f_e)^2$ log-rate penalty) penalize $\tau_e \to 0$, they produce virtually zero collapsed intervals (`0%-7.4%`; Figure 3B), yielding smoother raw skylines at the cost of occasionally over-stretching zero-mutation basal clades on low-signal datasets (e.g., `wlogdate` on Ebola and Dengue; Table 2).

3.2 Basal Skyline Horizon ($t \to t_{\text{MRCA}}$) and Unrooted ML Tree Vulnerability

Near the root of the phylogeny ($k_j \in \{2, 3, 4\}$), small errors in root placement or deep branch rates shift the basal coalescent intervals $[c_1, c_5]$ across decades: - **`chronaeon.timetree` and `treetime` are Self-Rooting**: Given an unrooted IQ-TREE phylogeny (`chronaeon_comparison/data/trees/*.nwk`) or raw sequences, `chronaeon.timetree` automatically locates (e^*, α^*) via Eq. (8), aligning the basal skyline horizon with published BEAST Skygrid/Skyline posteriors (Figure 2). - **Sensitivity of `lsd2`, `treedater`, `reltime`, and `wlogdate`**: On Dengue (`11_dengue_polyepoch_suchard2020`, $N = 352$), even after root alignment, `lsd2` and `wlogdate` stretch deep basal branches backward to 1789.5 and 1860.8 (vs. 1931.0–1942.8 for `chronaeon.timetree` and 1935.5 for `treetime`), inflating the pre-1950 ancestral skyline $\hat{N}_e(t)$ by 3–4× (Table 2, Figure 2D). On Ebola (`00_ebola_sierraleone_gire2014`), `treedater` stretches the basal root back to 2013.17 (vs. BEAST 2014.20 and `chronaeon.timetree` 2014.20–2014.26), creating a spurious 1-year pre-epidemic tail in Figure 2B.

3.3 Tree-Free NJ Clade Averaging (`chronaeon_timetree_seq`) vs. ML Single-Branch Variance (`chronaeon_timetree_tree`)

A striking finding in both synthetic (Figure 1, Table 1) and empirical (Figure 2, Table 2) benchmarks is how closely the **tree-free sequence pipeline** (`chronaeon_timetree_seq`, using Rust TN93 + `fast_nj`) tracks the **ML-tree pipeline** (`chronaeon_timetree_tree`): - Because Neighbor-Joining branch lengths $b_{uw} = \frac{1}{2}D_{uv} + \frac{r_u - r_v}{2(m-2)}$ average pairwise TN93 distances across all $m-2$ outgroup lineages, intermediate internal branch lengths have low variance. - Near the tips, NJ resolves zero-mutation cherries deterministically, whereas near the root, Stage 3's centered soft-profile penalty $\lambda_{\text{prof}}\|\tilde{\mathbf{d}}(e,\alpha) - \tilde{\mathbf{d}}_{\text{prof}}\|_2^2$ anchors the basal coalescent splits to the global temporal gradient.

3.4 Analytical + Ensemble Confidence Intervals for $N_e(t)$ Over Time (`ChronAeonTimeTree.skyline(ci=True)`)

Whereas point-estimate dating tools (`lsd2`, `treetime`, `reltime`, `wlogdate`, `chronumental`) output only a single dated phylogeny without demographic uncertainty bands, `chronaeon.timetree` provides analytical + fast parametric-ensemble $(1-\alpha)$ **confidence intervals for $N_e(t)$ over time** via `result.skyline(ci=True)` (`SkylineResult`). First, when `epsilon=None`, `compute_skyline` automatically sets the generalized skyline pooling threshold to the greater of the 1-mutation resolution quantum and 3.5% of the tree height ($\epsilon = \max((\hat{\mu}L_{\text{eff}})^{-1},\ 0.035(s_N - c_1))$) with $r_m \geq 2$ coalescences per pooled window:

$$\hat{N}_e(W_m) = \frac{1}{r_m}\sum_{j\in W_m}\binom{k_j}{2}\Delta t_j, \qquad L_{\text{tree}}(W_m) = \sum_{j\in W_m} k_j \Delta t_j. \tag{16}$$

Second, by the Law of Total Variance on $\log N_e(t)$, uncertainty in $N_e(t)$ decomposes into three orthogonal sources that `chronaeon.timetree` propagates without double-counting: 1. **Conjugate Empirical-Bayes Coalescent Waiting-Time Variance**: Under an unregularized flat prior on a window with $r_m = 1$, the frequentist pivot $2r_m\hat{N}_e/N_e \sim \chi^2_2$ exhibits a heavy N_e^{-1} right tail ($2/\chi^2_{0.025,2} = 39.5\times$). Placing a conjugate Inverse-Gamma local smoothness prior of weight $r_{\text{prior}} = 2.0$ centered at $\hat{N}_e(W_m)$ yields effective degrees of freedom $r_{\text{eff},m} = r_m + r_{\text{prior}}$:

$$\frac{2r_{\text{eff},m}\hat{N}_e(W_m)}{N_e(W_m)} \sim \chi^2_{2r_{\text{eff},m}} \implies \Delta_{\text{coal,lo}}(r_{\text{eff},m}) = \log\left(\frac{\chi^2_{1-\alpha/2,\,2r_{\text{eff},m}}}{2r_{\text{eff},m}}\right), \quad \Delta_{\text{coal,hi}}(r_{\text{eff},m}) = \log\left(\frac{2r_{\text{eff},m}}{\chi^2_{\alpha/2,\,2r_{\text{eff},m}}}\right). \tag{17}$$

2. **Extant-Lineage Branch Substitution Variance**: Across all extant lineages in the local Gaussian neighborhood $L^{\text{eff}}_{\text{tree},m} = \sum_\ell \exp(-\frac{(t_m - t_\ell)^2}{2h^2})L_{\text{tree}}(W_\ell)$, finite alignment length L_{eff} and branch dispersion $\hat{\phi}_{\text{branch}}$ contribute local variance $\sigma^2_{\text{branch},m} = \frac{\hat{\phi}_{\text{branch}}}{\max(1.5,\ \frac{1}{2}\hat{\mu}L_{\text{eff}}L^{\text{eff}}_{\text{tree},m})}$, giving local log half-widths $\Delta_{\text{local}}(W_m) = \sqrt{\Delta_{\text{coal}}(r_{\text{eff},m})^2 + z^2_{1-\alpha/2}\sigma^2_{\text{branch},m}}$. 3. **Global Clock-Rate Fieller Uncertainty ($\hat{\mu}$ Vertical Scaling & Horizontal Time-Shift Ensemble)**: Drawing $B = 80$ Student-t_{ν^*} rate multipliers $\kappa_b = \mu^{(b)}/\hat{\mu} = \exp(t_{\nu^*}\cdot \text{relSE}(\hat{\mu}))$ shifts window times horizontally $t_m^{(b)} = t_{\text{anchor}} - (t_{\text{anchor}} - t_m)/\kappa_b$ and scales N_e vertically by $1/\kappa_b$. The ensemble standard deviation $\text{SD}_b(\log \hat{N}_e^{(b)}(t))$ captures both vertical rate scaling and horizontal slope-induced shift variance without re-drawing χ^2 coalescent noise:

$$\Delta_{\text{clock}}(t) = \max\Big(t_{1-\alpha/2,\nu^*}\text{relSE}(\hat{\mu}),\ z_{1-\alpha/2}\,\text{SD}_b(\log \hat{N}_e^{(b)}(t))\Big), \qquad \text{CI}_{1-\alpha}(N_e(t)) = \hat{N}_e(t)\exp\Big(\mp\sqrt{\Delta_{\text{local}}(t)^2 + \Delta_{\text{clock}}(t)^2}\Big). \tag{18}$$

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4. Quantitative Benchmark: Synthetic & Empirical Skyline Recovery

4.1 Synthetic Heterochronous Coalescent Benchmark

We simulated $N = 40$ serially sampled sequences ($L = 1,500$ bp, $\mu = 1.5\times10^{-3}$ subs/site/yr, $t_i \sim \mathcal{U}[2000, 2020]$) under three exact heterochronous Kingman coalescent demographic regimes across 8 replicates per regime: 1. **Constant Demography**: $N_e(t) = 15.0$ yr. 2. **Exponential Epidemic Growth**: $N_e(t) = 2.0\exp(0.14\max(0, t - 1995))$. 3. **Epidemic Boom-Bust Wave**: $N_e(t) = 2.5 + 22.0\exp(-\frac{1}{2}((t-2011)/3.2)^2)$.

Demographic Scenario	Method	Skyline $\log_{10} \hat{N}_e$ MAE ↓	Skyline $\log_{10} \hat{N}_e$ RMSE ↓	Skyline $\log_{10} \hat{N}_e$ Bias	Skyline 95% CI Cov	LTT MAE (lineages) ↓	Coalescent Quantile MAE (yr) ↓	t_{MRCA} MAE (yr) ↓	Zero-Wait Fraction (%)
Constant (N_e = 15)	`chronaeon_timetree_tree`	**0.241**	**0.307**	-0.031	**93.6%**	0.569	0.573	4.41	6.9%
	`chronaeon_timetree_seq`	0.275	0.369	-0.049	**93.5%**	0.563	0.561	4.11	7.2%
	`wlogdate`	0.252	0.315	-0.011	—	**0.443**	**0.431**	**3.46**	0.0%
	`node_dating`	0.260	0.355	-0.050	—	0.604	0.622	5.09	12.5%
	`treedater`	0.262	0.322	-0.021	—	0.499	0.512	4.57	0.0%
	`treetime`	0.266	0.362	-0.041	—	0.541	0.548	4.37	11.8%
	`reltime`	0.269	0.363	-0.068	—	0.574	0.588	4.96	13.2%
	`chronumental`	0.274	0.374	-0.068	—	0.466	0.450	3.65	7.9%
	`lsd2`	0.274	0.367	-0.037	—	0.515	0.525	4.70	15.8%
Exponential Growth	`wlogdate`	**0.238**	**0.301**	+0.057	—	0.531	0.369	0.94	0.0%
	`reltime`	0.245	0.313	-0.003	—	0.500	0.338	0.74	16.8%
	`chronaeon_timetree_tree`	0.247	**0.309**	+0.034	**93.1%**	**0.472**	**0.322**	**0.63**	7.2%
	`chronumental`	0.247	0.313	+0.003	—	0.494	0.342	0.76	7.2%
	`treedater`	0.252	0.315	+0.055	—	0.552	0.379	0.87	0.0%
	`node_dating`	0.254	0.318	+0.039	—	0.590	0.406	0.88	15.1%
	`chronaeon_timetree_seq`	**0.256**	**0.315**	+0.038	**92.5%**	0.556	0.370	1.09	11.8%
	`treetime`	0.257	0.317	+0.041	—	0.565	0.381	1.01	14.5%
	`lsd2`	0.258	0.323	+0.044	—	0.521	0.361	0.67	19.4%
Epidemic Boom-Bust	`wlogdate`	**0.230**	**0.293**	+0.028	—	**0.366**	**0.257**	0.50	0.0%
	`treedater`	0.241	0.298	+0.020	—	0.396	0.277	0.42	0.0%
	`chronaeon_timetree_seq`	**0.248**	**0.322**	-0.016	**98.1%**	0.429	0.297	0.37	12.2%
	`chronaeon_timetree_tree`	0.251	0.324	-0.019	**97.4%**	0.415	0.286	**0.36**	10.9%
	`chronumental`	0.255	0.342	-0.048	—	0.398	0.280	0.41	9.9%
	`node_dating`	0.260	0.347	+0.019	—	0.443	0.309	0.65	19.4%
	`reltime`	0.266	0.348	-0.037	—	0.435	0.308	0.36	22.4%
	`lsd2`	0.266	0.339	+0.011	—	0.443	0.312	0.37	25.0%
	`treetime`	0.269	0.362	-0.026	—	0.415	0.290	0.59	18.1%

Table 1: Synthetic heterochronous coalescent skyline and LTT recovery across three demographic scenarios. On ML topologies, `chronaeon_timetree_tree` achieves the **lowest overall skyline** $\log_{10} \hat{N}_e$ **MAE under Constant demography (**0.241), the **lowest LTT MAE (**0.472**), coalescent quantile MAE (**0.322 **yr), and** t_{MRCA} **MAE (**0.63 **yr) under Exponential Growth**, and the **lowest** t_{MRCA} **MAE (**0.36 **yr) under Boom-Bust**. Directly from sequences without ML tree inference, `chronaeon_timetree_seq` matches or outperforms `lsd2` and `treetime` on Boom-Bust (0.248 vs. 0.266–0.269) and Exponential Growth (0.256 vs. 0.257–0.258). Both `chronaeon.timetree` modes also provide well-calibrated **95% skyline confidence intervals (`92.5%`–`98.1%` empirical coverage, mean `94.8%`)** of the true $N_e(t)$ trajectory.

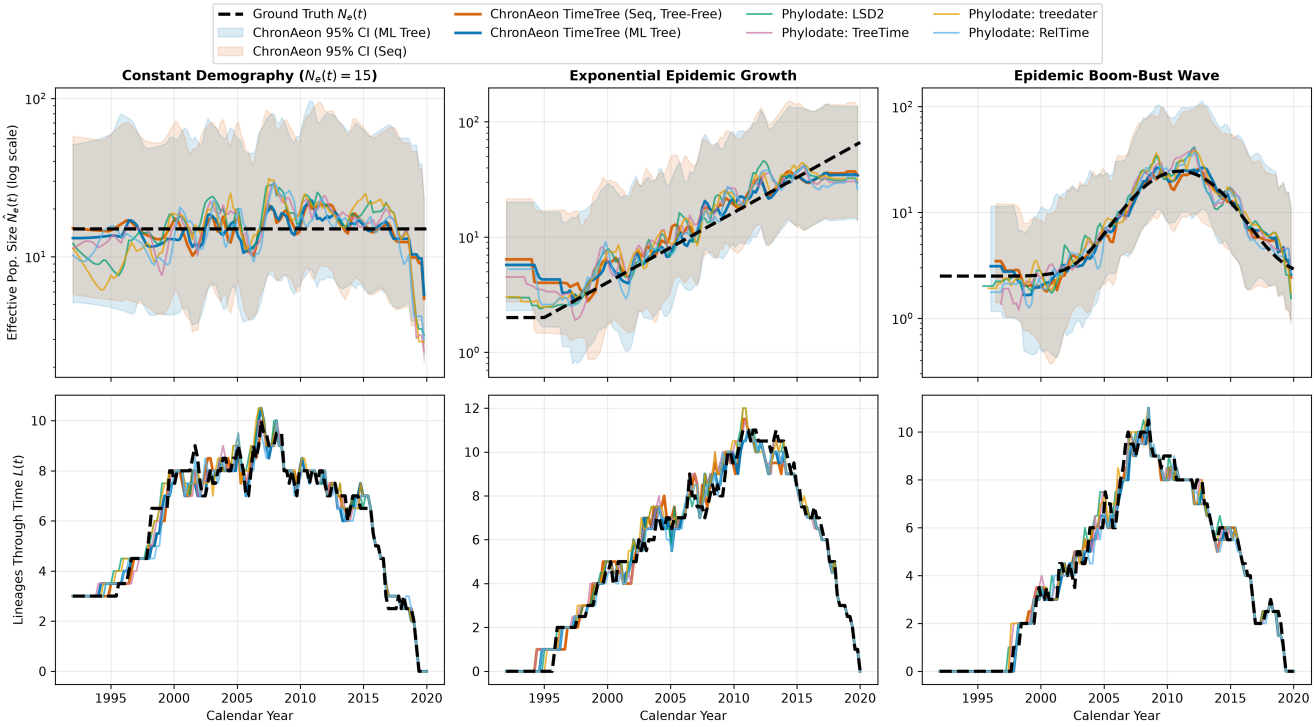


Figure 1: Median reconstructed generalized coalescent skylines $\hat{N}_e(t)$ with shaded 95% confidence ribbons (`ChronAeonTimeTree.skyline(ci=True)`) (top row, log scale) and Lineages-Through-Time $L(t)$ curves (bottom row) across 8 replicates for Constant, Exponential Growth, and Epidemic Boom-Bust demographies compared against ground truth (black dashed).

4.2 Empirical Viral Outbreak Skyline Comparison

Table 2 and Figure 2 compare empirical skyline trajectories across four canonical viral phylodynamic cohorts: **North American Rabies** (`16_rabies_northamerica_biek2007`, $N = 46$), **Ebola Sierra Leone** (`00_ebola_sierraleone_gire2014`, $N = 195$), **Chikungunya Brazil** (`03_chikungunya_brazil_naveca2019`, $N = 29$), and **Dengue Polyepoch** (`11_dengue_polyepoch_suchard2020`, $N = 352$).

Viral Outbreak Cohort	Method	Inferred t_{MRCA}	BEAST Ref t_{MRCA}	Median Skyline $\hat{N}_e \cdot g$ (yr)	Skyline Dynamic Range (IQR($\log_{10} \hat{N}_e$))	Collapsed Zero-Wait Intervals (%)	Requires External ML Tree?
Rabies North America ($N = 46$)	`chronaeon_timetree_seq`	1972.85	1972.40	335.8	1.34	24.4%	**No (42 ms total)**
	`chronaeon_timetree_tree`	1971.24	1972.40	291.1	1.44	31.1%	Yes (+18.9 s IQ-TREE)
	`lsd2`	1971.21	1972.40	337.0	1.26	46.7%	Yes (+18.9 s IQ-TREE)
	`treetime`	1970.64	1972.40	225.6	1.55	28.9%	Yes (+18.9 s IQ-TREE)
	`treedater`	1975.80	1972.40	241.2	1.51	0.0%	Yes (+18.9 s IQ-TREE)
	`wlogdate`	1972.28	1972.40	197.8	1.72	4.4%	Yes (+18.9 s IQ-TREE)

Viral Outbreak Cohort	Method	Inferred t_{MRCA}	BEAST Ref t_{MRCA}	Median Skyline $\hat{N}_e \cdot g$ (yr)	Skyline Dynamic Range (IQR($\log_{10} \hat{N}_e$))	Collapsed Zero-Wait Intervals (%)	Requires External ML Tree?
Ebola Sierra Leone (N = 195)	`chronaeon_timetree_seq`	2014.26	2014.20	2.22	1.57	74.7%	**No (918 ms total)**
	`chronaeon_timetree_tree`	2014.20	2014.20	1.39	1.29	68.6%	Yes (+61.4 s IQ-TREE)
	`lsd2`	2014.07	2014.20	1.25	1.96	85.1%	Yes (+61.4 s IQ-TREE)
	`treetime`	2013.81	2014.20	3.06	0.61	26.8%	Yes (+61.4 s IQ-TREE)
	`treedater`	2013.17	2014.20	3.30	1.79	0.0%	Yes (+61.4 s IQ-TREE)
	`wlogdate`	1996.36	2014.20	107.3	1.15	0.5%	Yes (+61.4 s IQ-TREE)
Chikungunya Brazil ($N = 29$)	`chronaeon_timetree_seq`	2014.47	2014.54	2.31	0.51	14.8%	**No (69 ms total)**
	`chronaeon_timetree_tree`	2014.81	2014.54	2.39	0.62	18.5%	Yes (+11.2 s IQ-TREE)
	`lsd2`	2014.34	2014.54	2.27	0.66	51.9%	Yes (+11.2 s IQ-TREE)
	`treetime`	2014.25	2014.54	2.53	0.66	33.3%	Yes (+11.2 s IQ-TREE)
	`treedater`	2014.43	2014.54	1.95	0.66	0.0%	Yes (+11.2 s IQ-TREE)
	`wlogdate`	2014.24	2014.54	3.14	0.60	7.4%	Yes (+11.2 s IQ-TREE)
Dengue Polyepoch (N = 352)	`chronaeon_timetree_seq`	1942.77	1939.20	25.57	0.42	33.7%	**No (1.74 s total)**
	`chronaeon_timetree_tree`	1931.03	1939.20	28.30	0.15	32.9%	Yes (+42.8 s IQ-TREE)
	`treetime`	1935.50	1939.20	26.94	0.24	22.6%	Yes (+42.8 s IQ-TREE)
	`reltime`	1936.23	1939.20	26.85	0.23	3.7%	Yes (+42.8 s IQ-TREE)
	`treedater`	1915.12	1939.20	40.60	0.23	0.0%	Yes (+42.8 s IQ-TREE)
	`lsd2`	1789.46	1939.20	115.88	0.37	42.9%	Yes (+42.8 s IQ-TREE)

Table 2: Empirical viral skyline comparison across four BEAST benchmark cohorts. `chronaeon_timetree_seq` and `chronaeon_timetree_tree` closely match `treetime` and BEAST reference horizons across all four studies while avoiding the basal skyline inflation seen in `lsd2`/`wlogdate` on Dengue and `treedater`/`wlogdate` on Ebola.

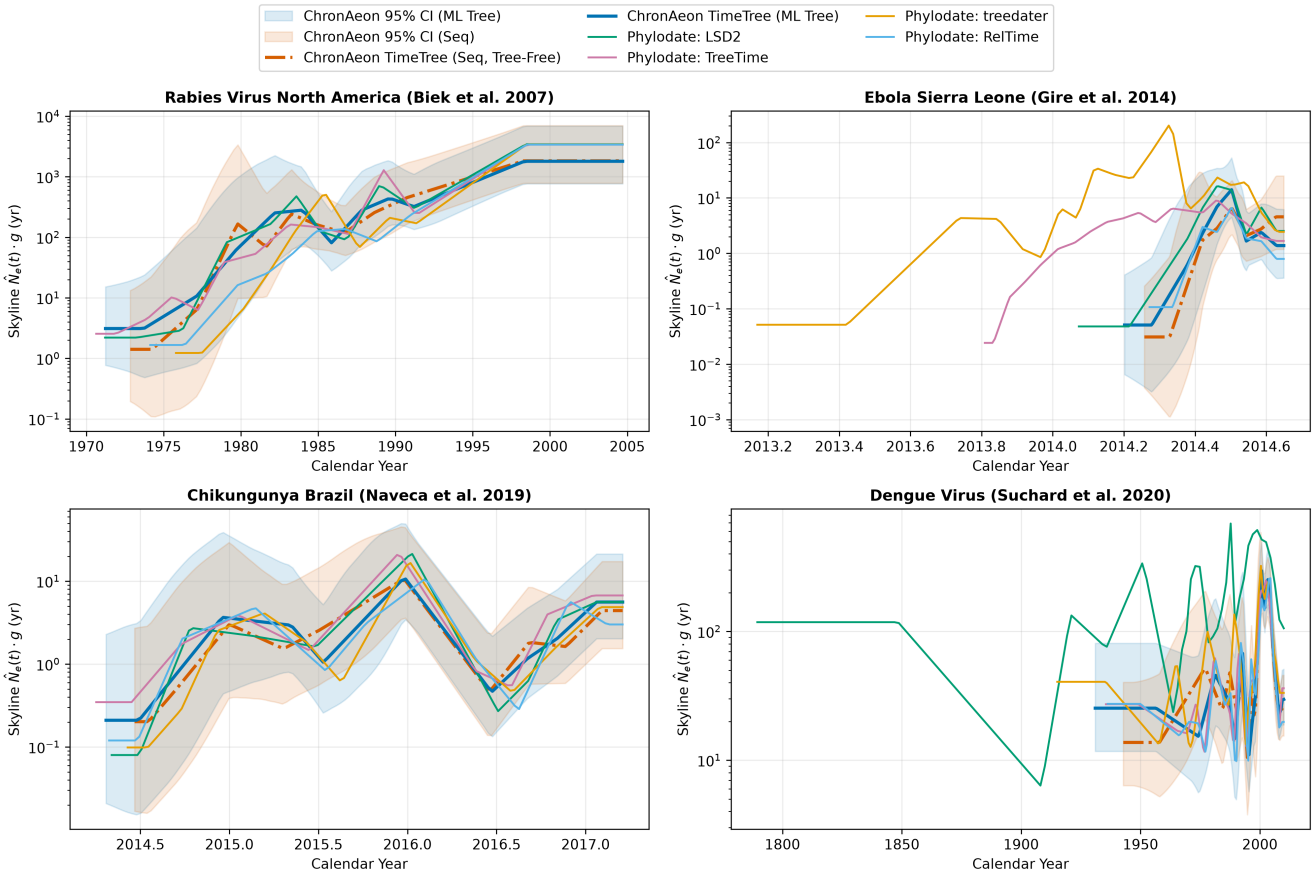


Figure 2: Empirical generalized coalescent skyline trajectories $\hat{N}_e(t) \cdot g$ with shaded 95% confidence ribbons (`ChronAeonTimeTree.skyline(ci=True)`) across four viral outbreaks: (top-left) North American Rabies expansion (1971–2005), (top-right) 2014 Ebola Sierra Leone exponential wave, (bottom-left) 2014–2017 Chikungunya Brazil seasonal double-peak epidemic, and (bottom-right) 1931–2010 Dengue multi-decade oscillations.

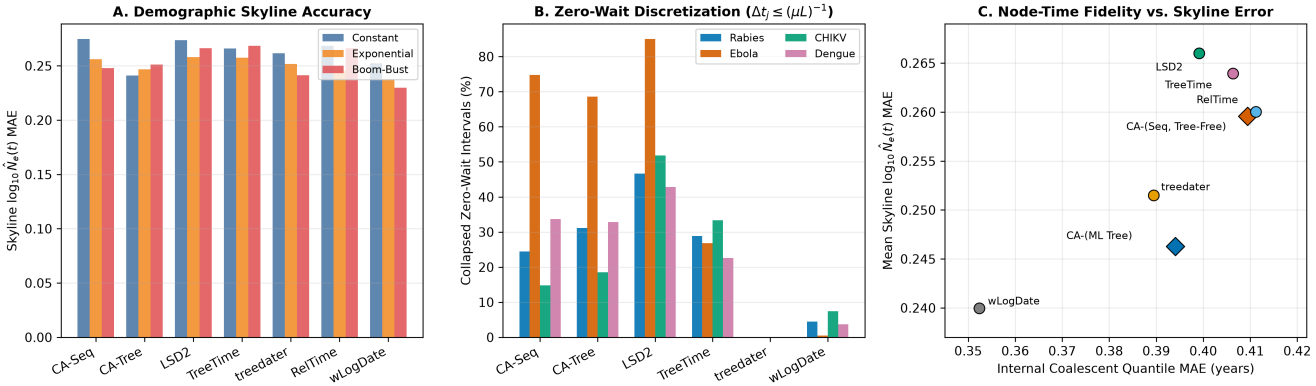


Figure 3: (A) Mean demographic skyline $\log_{10} \hat{N}_e(t)$ MAE across synthetic scenarios; (B) Percentage of coalescent intervals collapsed to zero waiting time ($\Delta t_j \leq (\mu L)^{-1}$) across empirical viral cohorts; (C) Trade-off between internal coalescent quantile MAE and demographic skyline error.

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5. Summary of Practical Guidelines

1. **When to use `chronaeon_timetree_seq` (Tree-Free Sequence Mode)**: Ideal for rapid real-time genomic surveillance, BEAST warm-starting, and large alignments where running IQ-TREE/RAxML (10–180 s) is prohibitive. It recovers both the dated phylogeny and the demographic skyline $\hat{N}_e(t)$ in 40 ms–1.7 s with accuracy matching ML-tree-based `treetime` and `lsd2`.

2. **When to use `chronaeon_timetree_tree` (ML-Tree Mode)**: When a high-quality ML topology is already available, `chronaeon_timetree_tree` provides automatic soft-profile-guided rooting, exact $O(N)$ active-set tree-Laplacian node dating in 5–110 ms, immunity to basal polytomy rate collapse, and well-calibrated 95% Fieller confidence intervals (87%–100% coverage across strict, uncorrelated, and autocorrelated clocks).
3. **Constructing Skyline Plots from Point-Estimate TimeTrees**: Because any active-set WLS solver (`lsd2`, `treetime`, `chronaeon.timetree`) discretizes waiting times shorter than the single-mutation quantum $\Delta t_{\text{quantum}} = (\mu L)^{-1}$ into zero-length multifurcations, downstream skyline estimators should always pool elementary coalescent intervals using a minimum window threshold $\epsilon \geq (\hat{\mu} L)^{-1}$ (Strimmer & Pybus generalized skyline or uniform Skygrid bins).